



THE IMPACT OF EDUCATION ON COVID-19 DEATHS

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Abstract

Covid-19 threats global since 2020, there are still many hidden information about how demographic environment interacts with the Covid-19 disease. By joining the 2020 US Census and Covid-19 datasets in the database, we were able to analyze many attributes and found that the Covid-19 deaths is significantly related to the education level.

Datasets and Methods

1. Two different datasets: 2020 U.S. Census and Covid-19 stored in 2 different MySQL tables.
2. 2935 US counties and Covid-19 data from 1/21/2020 to 12/31/2021 were used for the analysis.
3. Use google charts to visualize the datasets and results.
4. Use SQL to join these two tables and set the condition to eliminate the non-updating data and small population area.

Platforms

- Linux system, MySQL Database
- Code: HTML, JavaScript, PHP, Python
- API: Google Charts

References

- 2020 US Census dataset: <https://simplemaps.com/>
- Covid-19 dataset: <https://covid.ourworldindata.org/>

Experiments and Results

# state	county	deaths 2021_12	cases 2021_12	college_or_above	less_high school	death rate
Texas	Sabine	77	986	17.6	15.3	7.8093
Georgia	Hancock	83	1210	8.6	25.8	6.8595
Texas	Knox	22	390	20.1	18.1	5.641
South Dakota	Jerauld	19	337	22.5	10.6	5.638
Georgia	Candler	90	1619	13	22.7	5.559
Texas	Newton	54	987	9.8	16.6	5.4711
Texas	Haskell	37	686	11.9	29.7	5.3936
Texas	Dickens	15	280	18.5	15.6	5.3571
Georgia	Twiggs	62	1168	11.8	25.8	5.3082
Georgia	Jenkins	63	1210	7.8	20.7	5.2066

Join two table using MySQL workbench.

Column description:

cases2021_12: accumulation the total cases of Covid-19 from 01/21/2020 to 12/31/2021

population: population in the county

deaths2021_12: accumulation the total death cases of Covid-19 from 01/21/2020 to 12/31/2021

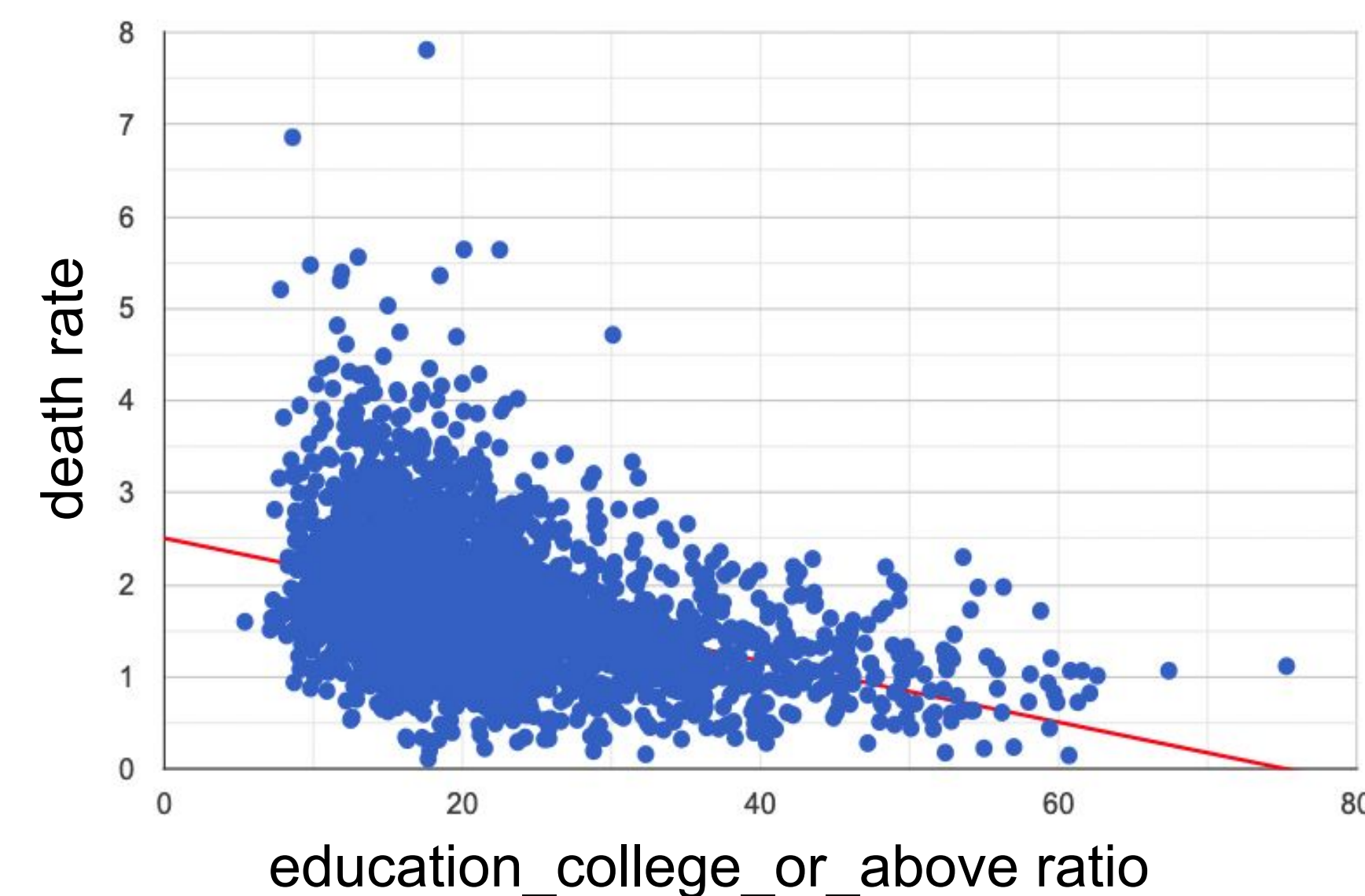
college_or_above: The percentage of residents with at least a 4-year degree

less_highschool: The percentage of residents with less than a high school education

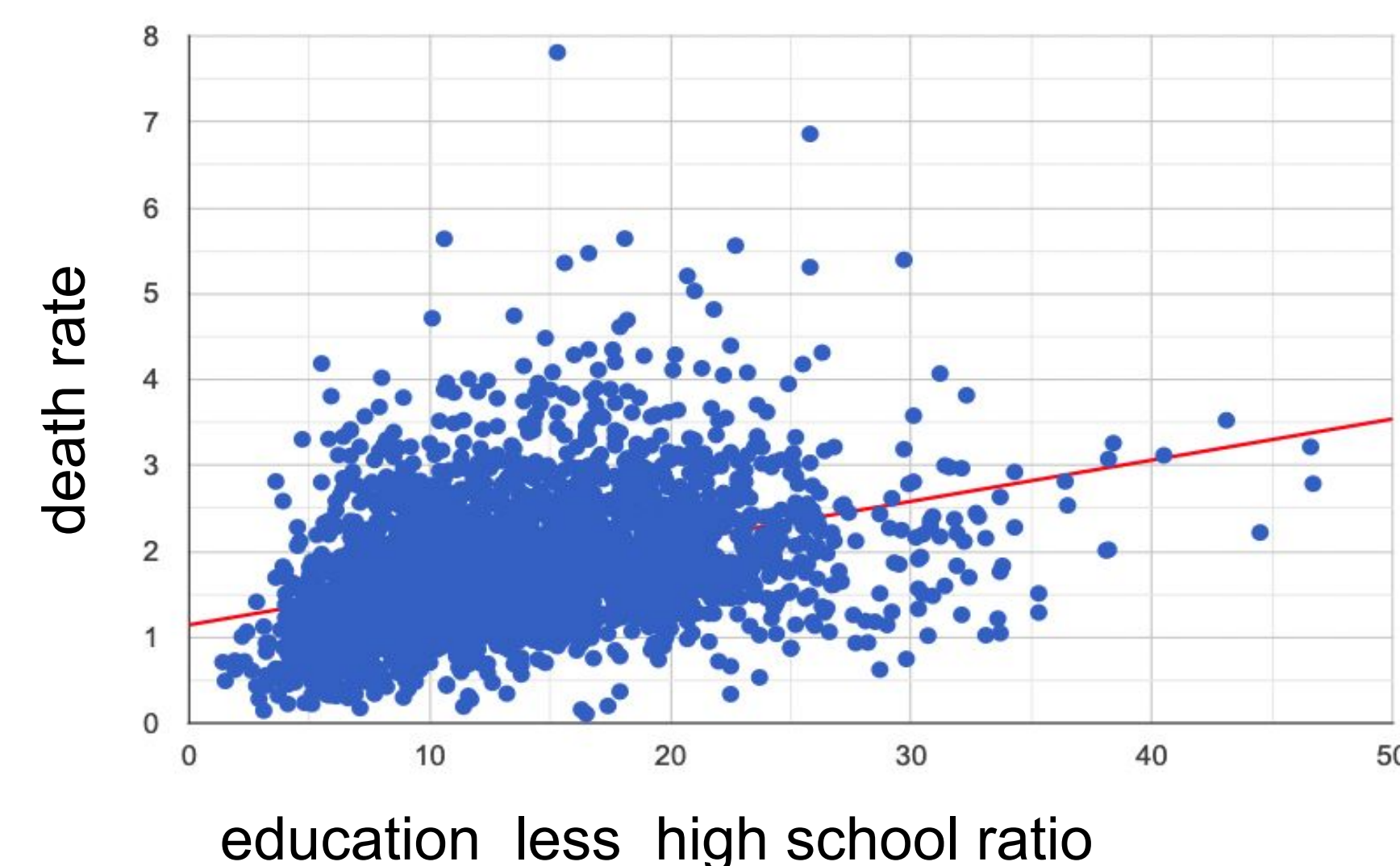
death_rate=death numbers/ case numbers

The top 10 high death rate counties are mostly at Texas and Georgia.

education_college_or_above ratio VS death rate



education_less_high school ratio VS death rate



A PHP program is access the data from MySQL database and display the results using Google Charts API. The chart shows the relationship between death rate and ratio of education level above college.

The graph implies that the higher ratio of education above college has the lower death rate. The correlation between those two variables is -0.415, which is considered a weak relation.

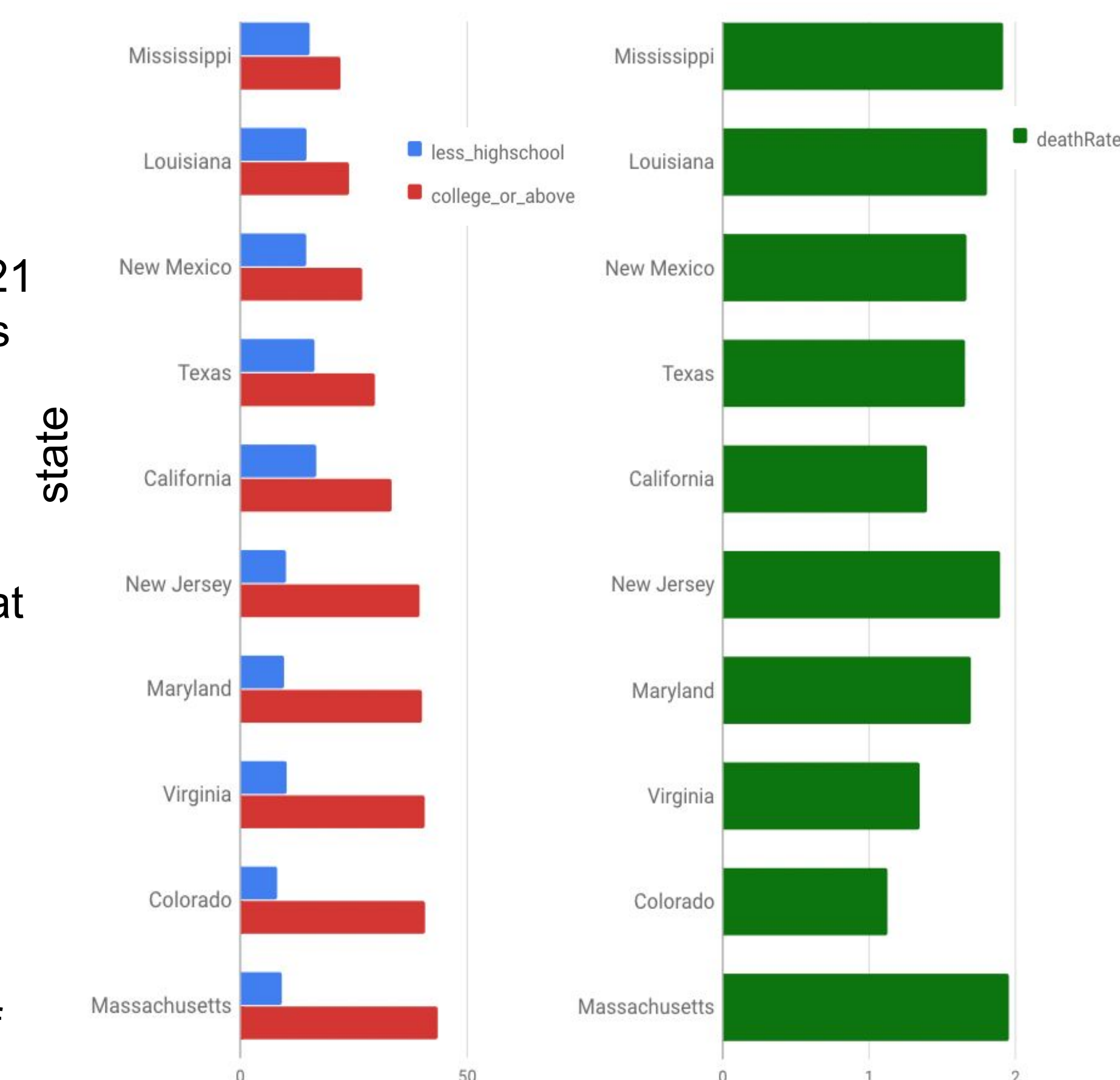
Summary

We join two dataset and analysis the potential attribute which may impact on COVID-19 death rates.

By using scatter plot, we found that there is a negative correlation between the ratio of education of college above.

Also, we list 5 counties with higher percentages of education level with college or above, and 5 counties with higher percentages of education level with less than high school. We found there is a decreasing tendency between these two factors.

education level distribution and death rate



Counties with highest percentage of education above college which are NJ, MD, VA, CO, MA. Counties with highest percentage of education less than high school which are MS, LA, NM, TX, CA.

We calculate the state statistics from the county data.

By comparing two graphs, we could see there is a decreasing tendency on death rate with ratio of education_college_or_above, but there are outliers in our results.

Acknowledgement

This research was funded by the Students Partnering with Faculty (SpF) grant supported by the Office of Research and Sponsored Programs (ORSP) at Kean University. We really appreciate Kean University's support.